

GRAVES.

The French Space Surveillance System

According to the American space catalog, more than 9,000 satellites or objects larger than ten centimeters are orbiting the Earth. Many of these overfly France daily, constituting a potential threat for the country. Until now, only the Americans and Russians have had an operational space surveillance system. In the early nineties, Onera proposed an independent system dubbed "Graves" for this, to watch satellites in low orbit (altitude less than 1,000 km).

The Graves system was developed under DGA contract and consists of a specific radar combined with an automatic processing system that creates and updates a database of the orbital parameters for the satellites it detects.

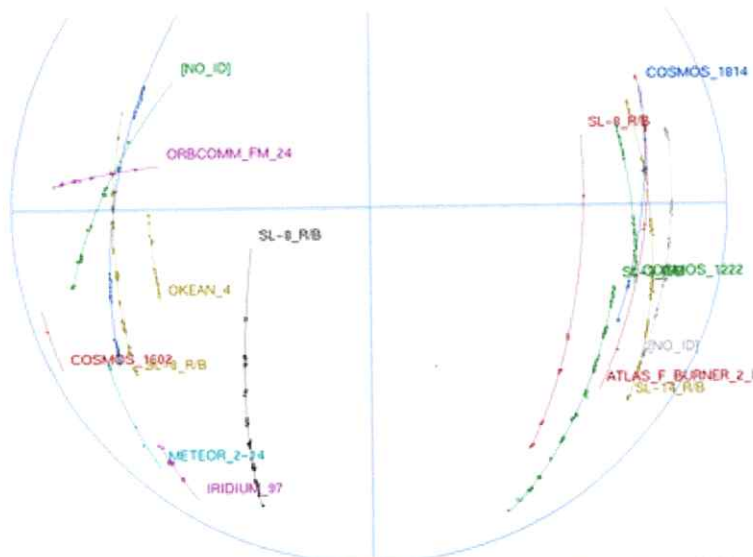
The Graves radar was specifically designed for space surveillance in cooperation among DEMR and DPRS specialists. The priority objective was to design a low-cost system, both in terms of development and maintenance. This led to an original bistatic radar concept with electronic scanning and continuous VHF band emission. The reception system is based on Doppler detection, and uses an innovative beam formation technique, by calculation. Using very conventional technology subassemblies, making it reliable and maintainable, radar performance is based on high-level signal processing requiring all the power of a dedicated real-time computer.



Graves radar emission site.

Downstream of the radar, a major effort was made to develop programs to convert the raw measurements into an orbital parameter database. This was a very ambitious technical challenge because, contrary to the American system that uses many sensors distributed around the globe, the tools developed for the Graves system only process the measurements from a single sensor.

After 15 years of work, the Graves system was delivered to its end user, the Air force, in November 2005. Operational since then, it has been keeping a database of some 2,000 satellites up to date. For a few dozen "sensitive" satellites, the U.S. does not publish the corresponding orbital elements.



2004-12-02 12:07:26.0 s=16 m=1 16/ 0/0.933 t=600 d=300 x=30 r=300 n=10 i=3 j=5 u=-0.1416667 v=-1.0729167 STOP

Tracks of satellites identified by orbital processing.